Maximum Likelihood Estimation and Model Selection for Stationary Renewable Energy Output Data

Guided walk-through. The heavy computations live in `src/` and are run by `main.py`; this notebook re-runs the key steps interactively and displays the saved results. Run `python main.py` first so that `results/` exists.

In [1]:
```
import sys, json, warnings
from pathlib import Path
ROOT = Path.cwd()
if not (ROOT / 'src').exists():
    ROOT = ROOT.parent
sys.path.insert(0, str(ROOT))
warnings.filterwarnings('ignore')
import numpy as np, pandas as pd
from IPython.display import Image, display
from src import config, preprocessing as pp, stationarity as st, mle_models as mm, model_selection as ms
pd.set_option('display.width', 200); pd.set_option('display.max_columns', 30)
```

1. Data: load, validate, aggregate

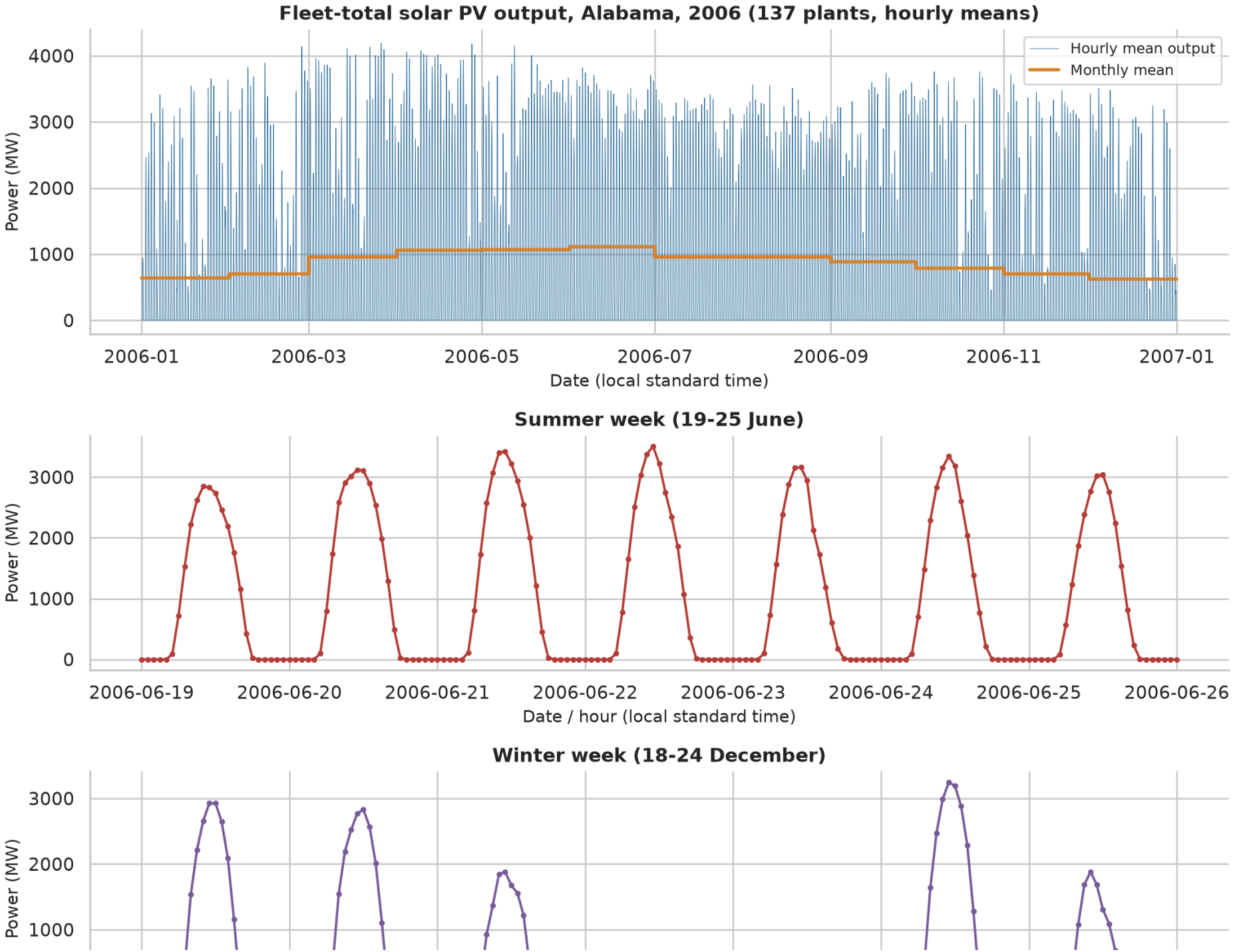
In [2]:
```
matrix = pp.load_raw_matrix()
index10 = pp.build_timestamps(len(matrix))
print(json.dumps(pp.validate_raw(matrix, index10), indent=1))
hourly = pp.to_hourly(pp.aggregate_fleet(matrix, index10))
frame = pp.build_analysis_frame(hourly)
print(pp.summarize_missing_and_unusual(frame))
frame.head(12)
```

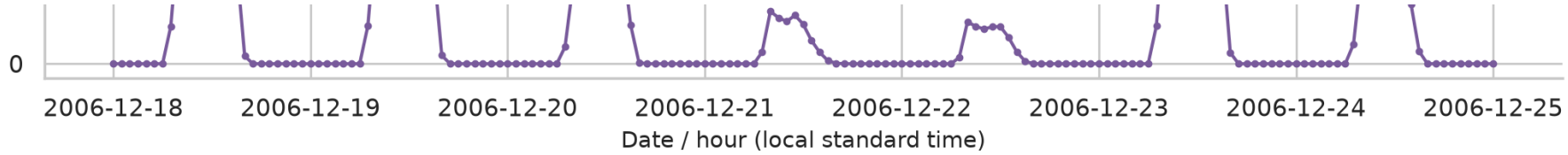
```
{
 "n_rows": 52560,
 "n_plants": 137,
 "expected_rows": 52560,
 "rows_match_expected": true,
 "plants_match_expected": true,
 "n_missing": 0,
 "n_non_finite": 0,
 "n_negative": 0,
 "n_duplicate_timestamps": 0,
 "n_distinct_time_steps": 1,
 "time_step_minutes": 10.0,
 "n_plants_always_zero": 0,
 "plant_max_mw": 88.9,
 "first_timestamp": "2006-01-01 00:00:00",
 "last_timestamp": "2006-12-31 23:50:00"
}
{'n_hours': 8760, 'n_hours_exact_zero': 4076, 'share_hours_exact_zero': 0.46529680365296805, 'n_daylight_hours': 3848,
'n_daylight_zero_output': 2, 'n_in_sample': 3846, 'n_unusual_flagged': 15, 'n_index_above_1_20': 1, 'n_index_below_0_0
5': 14, 'n_index_above_1': 233, 'max_index': 1.2545650917300142, 'envelope_threshold_mw': 413.2328333333333}
```


Out[2]:

	hourly_total_mw	envelope_mw	in_daylight	clearsky_index	in_sample	unusual	hour	day_of_year	month	day_index
2006-01-01 00:00:00	0.000000	0.000000	False	NaN	False	False	0	1	1	0
2006-01-01 01:00:00	0.000000	0.000000	False	NaN	False	False	1	1	1	0
2006-01-01 02:00:00	0.000000	0.000000	False	NaN	False	False	2	1	1	0
2006-01-01 03:00:00	0.000000	0.000000	False	NaN	False	False	3	1	1	0
2006-01-01 04:00:00	0.000000	0.000000	False	NaN	False	False	4	1	1	0
2006-01-01 05:00:00	0.000000	0.000000	False	NaN	False	False	5	1	1	0
2006-01-01 06:00:00	0.000000	0.000000	False	NaN	False	False	6	1	1	0
2006-01-01 07:00:00	197.833333	444.748333	True	0.444821	True	False	7	1	1	0
2006-01-01 08:00:00	813.958333	1741.346667	True	0.467430	True	False	8	1	1	0
2006-01-01 09:00:00	883.716667	2646.875000	True	0.333872	True	False	9	1	1	0
2006-01-01 10:00:00	868.541667	3113.679167	True	0.278944	True	False	10	1	1	0
2006-01-01 11:00:00	946.558333	3265.725000	True	0.289846	True	False	11	1	1	0

In [3]:
```
display(Image(str(config.FIG_DIR / 'fig01_hourly_series_overview.png')))
```

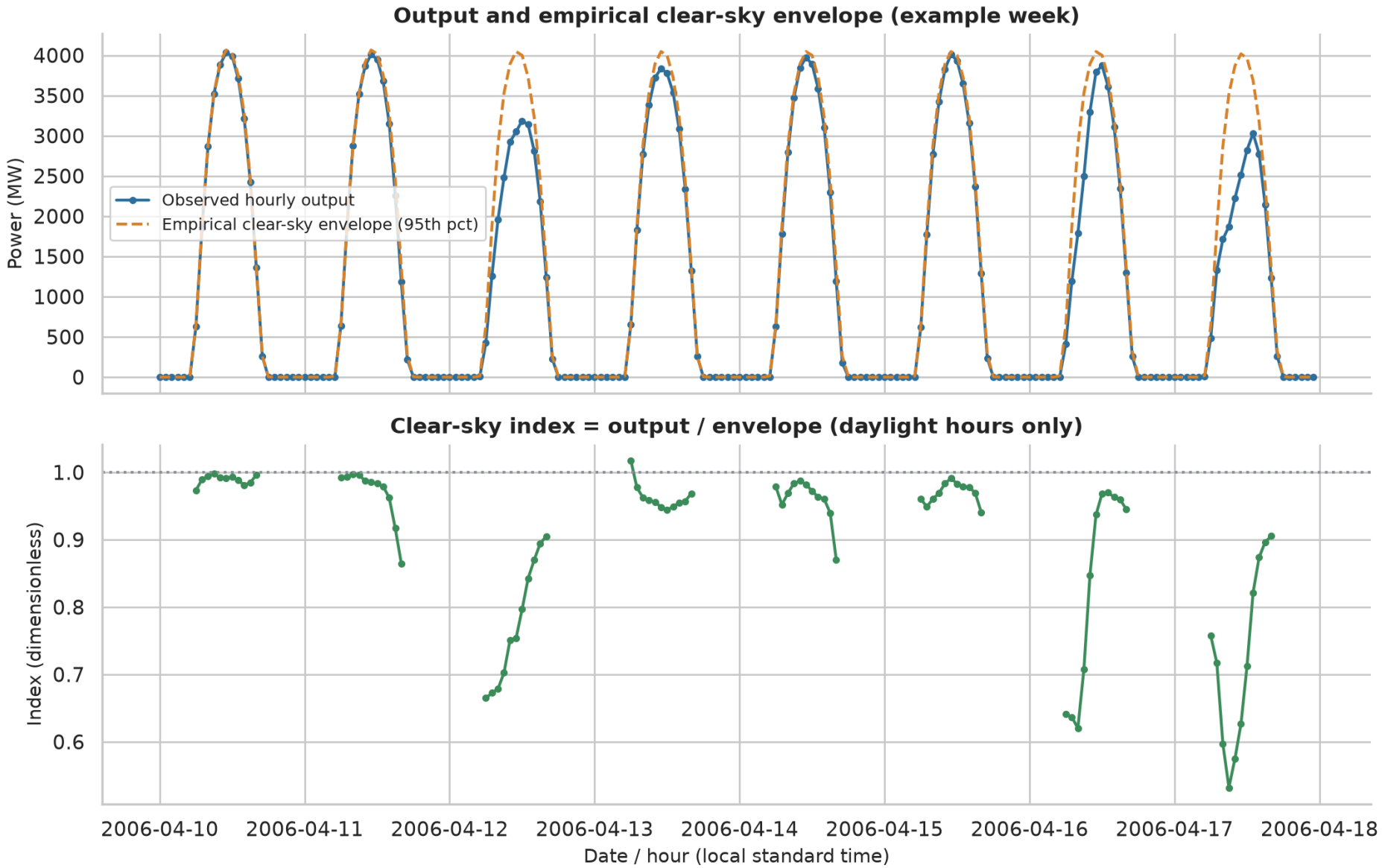

Fleet-total solar PV output, Alabama, 2006 (137 plants, hourly means)
Hourly mean output
Monthly mean
Power (MW)
4000
3000
2000
1000
0
2006-01
2006-03
2006-05
2006-07
2006-09
2006-11
2007-01
Date (local standard time)
Summer week (19-25 June)
Power (MW)
3000
2000
1000
0
2006-06-19
2006-06-20
2006-06-21
2006-06-22
2006-06-23
2006-06-24
2006-06-25
2006-06-26
Date / hour (local standard time)
Winter week (18-24 December)
Power (MW)
3000
2000
1000



2. Zeros and the clear-sky index

Night-time zeros are structural, so the analysis series is the clear-sky index (output divided by an empirical clear-sky envelope) on daylight hours.

In [4]:
```
display(Image(str(config.FIG_DIR / 'fig04_clearsky_envelope_and_index.png')))
```



3. Stationarity

In [5]:
```
tests = pd.read_csv(config.RESULTS_DIR / 'stationarity_tests.csv')
tests[['series','test','specification','statistic','p_value','reject_H0','interpretation']]
```

Out[5]:

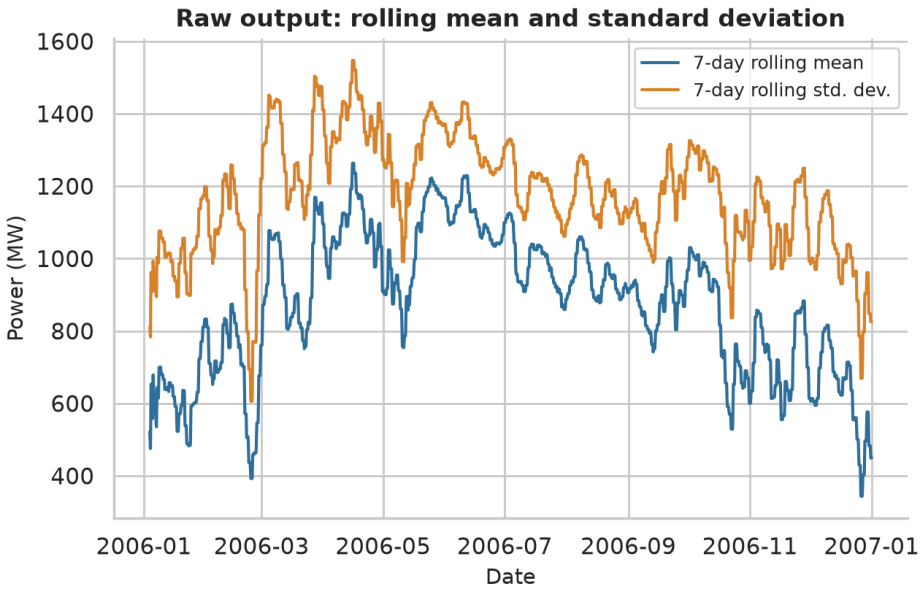
	series	test	specification	statistic	p_value	reject_H0	interpretation
0	Raw hourly output (MW)	ADF	regression='c', lags by AIC	-10.673600	4.106850e-19	True	Reject H0: evidence against a unit root.
1	Raw hourly output (MW)	KPSS	regression='c', bandwidth=auto (data-driven)	2.470810	1.000000e-02	True	Reject H0: evidence against level stationarity.
2	Raw hourly output (MW)	KPSS	regression='c', bandwidth=37 (long rule)	1.673910	1.000000e-02	True	Reject H0: evidence against level stationarity.
3	Raw hourly output (MW)	KPSS	regression='ct', bandwidth=auto (data-driven)	1.961100	1.000000e-02	True	Reject H0: evidence against trend stationarity.
4	Raw daily mean output (MW)	ADF	regression='c', lags by AIC	-2.846980	5.187030e-02	False	Fail to reject H0: a unit root cannot be ruled...
5	Raw daily mean output (MW)	KPSS	regression='c', bandwidth=auto (data-driven)	0.822788	1.000000e-02	True	Reject H0: evidence against level stationarity.
6	Raw daily mean output (MW)	KPSS	regression='c', bandwidth=17 (long rule)	0.533990	3.401120e-02	True	Reject H0: evidence against level stationarity.
7	Raw daily mean output (MW)	KPSS	regression='ct', bandwidth=auto (data-driven)	0.670481	1.000000e-02	True	Reject H0: evidence against trend stationarity.
8	Raw hourly, seasonal difference lag 24	ADF	regression='c', lags by AIC	-19.076700	0.000000e+00	True	Reject H0: evidence against a unit root.
9	Raw hourly, seasonal difference lag 24	KPSS	regression='c', bandwidth=auto (data-driven)	0.023280	1.000000e-01	False	Fail to reject H0: consistent with level stati...
10	Raw hourly, seasonal difference lag 24	KPSS	regression='c', bandwidth=37 (long rule)	0.022194	1.000000e-01	False	Fail to reject H0: consistent with level stati...
11	Raw hourly, seasonal difference lag 24	KPSS	regression='ct', bandwidth=auto (data-driven)	0.003458	1.000000e-01	False	Fail to reject H0: consistent with trend stati...
12	Clear-sky index, daylight hours (concatenated)	ADF	regression='c', lags by AIC	-12.399800	4.586980e-23	True	Reject H0: evidence against a unit root.
13	Clear-sky index, daylight hours (concatenated)	KPSS	regression='c', bandwidth=auto (data-driven)	0.830651	1.000000e-02	True	Reject H0: evidence against level stationarity.
14	Clear-sky index, daylight hours (concatenated)	KPSS	regression='c', bandwidth=30 (long rule)	0.872053	1.000000e-02	True	Reject H0: evidence against level stationarity.
15	Clear-sky index, daylight hours (concatenated)	KPSS	regression='ct', bandwidth=auto (data-driven)	0.712390	1.000000e-02	True	Reject H0: evidence against trend stationarity.
16	Clear-sky index at 12:00 (daily)	ADF	regression='c', lags by AIC	-6.272010	3.983140e-08	True	Reject H0: evidence against a unit root.
17	Clear-sky index at 12:00 (daily)	KPSS	regression='c', bandwidth=auto (data-driven)	0.600284	2.261050e-02	True	Reject H0: evidence against level stationarity.
18	Clear-sky index at 12:00 (daily)	KPSS	regression='c', bandwidth=17 (long rule)	0.379087	8.616920e-02	False	Fail to reject H0: consistent with level stati...
19	Clear-sky index at 12:00 (daily)	KPSS	regression='ct', bandwidth=auto (data-driven)	0.576729	1.000000e-02	True	Reject H0: evidence against trend stationarity.
20	Daily mean clear-sky index	ADF	regression='c', lags by AIC	-7.154910	3.072670e-10	True	Reject H0: evidence against a unit root.
21	Daily mean clear-sky index	KPSS	regression='c', bandwidth=auto (data-driven)	0.615372	2.123890e-02	True	Reject H0: evidence against level stationarity.
22	Daily mean clear-sky index	KPSS	regression='c', bandwidth=17 (long rule)	0.402264	7.617910e-02	False	Fail to reject H0: consistent with level stati...
23	Daily mean clear-sky index	KPSS	regression='ct', bandwidth=auto (data-driven)	0.549246	1.000000e-02	True	Reject H0: evidence against trend stationarity.
24	Clear-sky index minus annual harmonic	ADF	regression='c', lags by AIC	-13.584700	2.090840e-25	True	Reject H0: evidence against a unit root.
25	Clear-sky index minus annual harmonic	KPSS	regression='c', bandwidth=auto (data-driven)	0.026185	1.000000e-01	False	Fail to reject H0: consistent with level stati...
26	Clear-sky index minus annual harmonic	KPSS	regression='c', bandwidth=30 (long rule)	0.026357	1.000000e-01	False	Fail to reject H0: consistent with level stati...
27	Clear-sky index minus annual harmonic	KPSS	regression='ct', bandwidth=auto (data-driven)	0.021600	1.000000e-01	False	Fail to reject H0: consistent with trend stati...

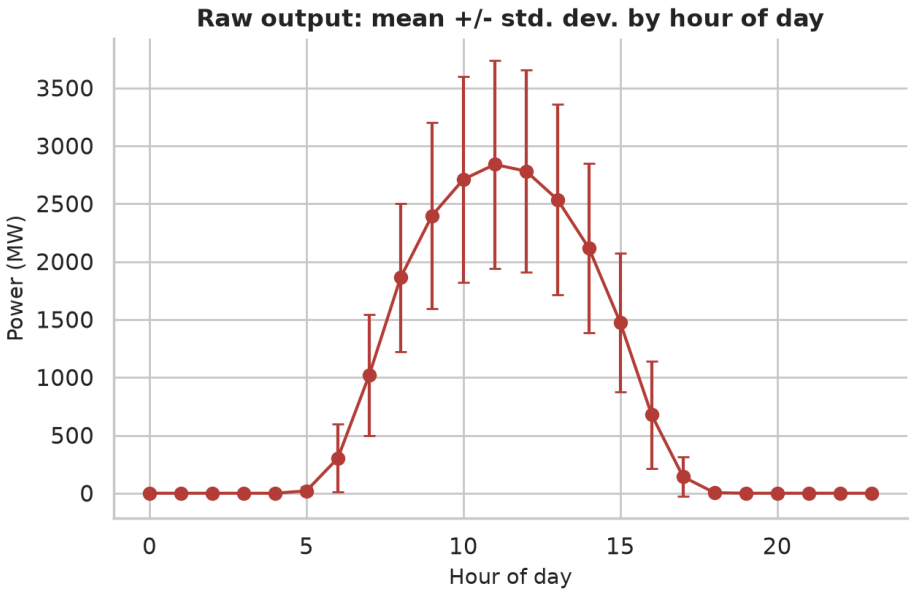
In [6]:
```
pd.read_csv(config.RESULTS_DIR / 'stationarity_verdicts.csv')
```

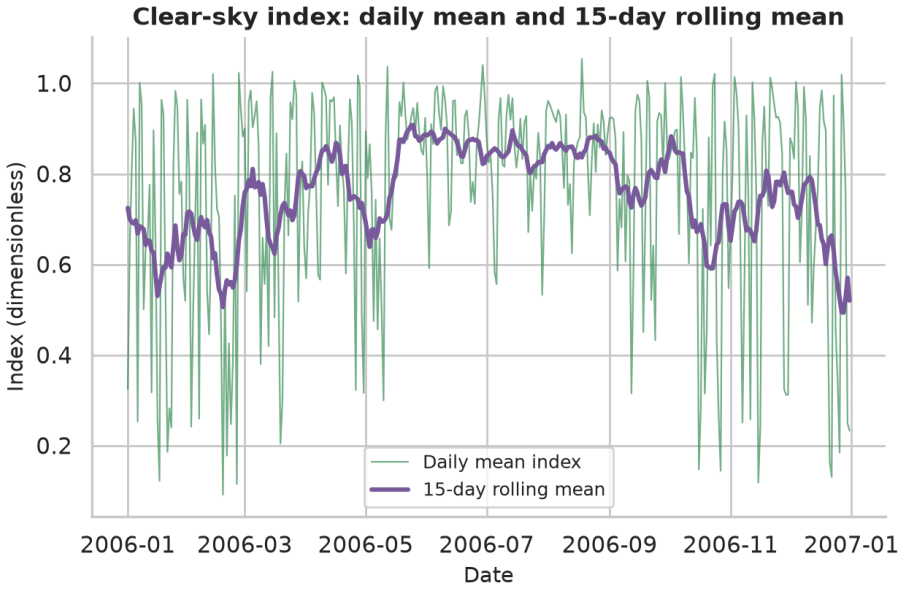
Out[6]:

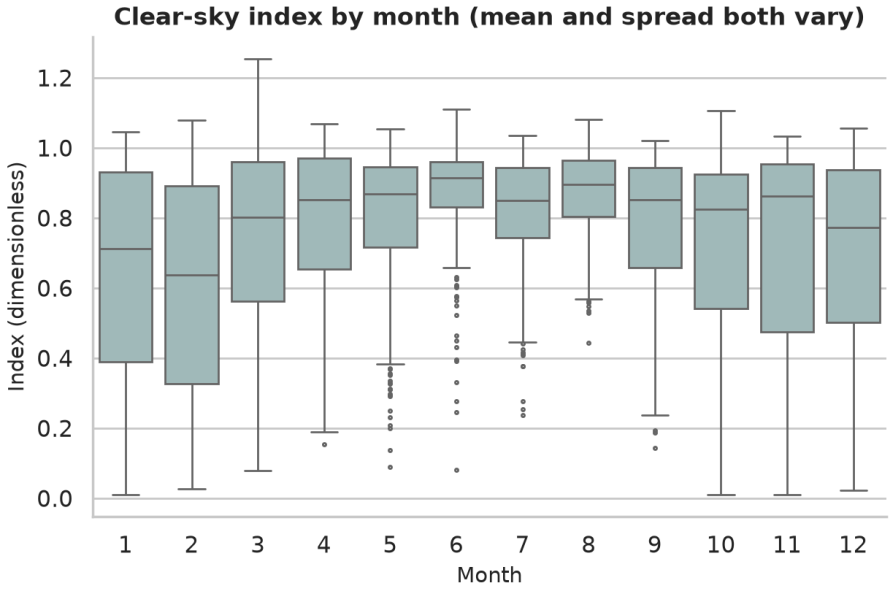
	series	adf_p	kpss_stat	kpss_p	verdict
0	Raw hourly output (MW)	4.106850e-19	2.470810	0.010000	Conflicting: no unit root but not level-statio...
1	Raw daily mean output (MW)	5.187030e-02	0.822788	0.010000	Non-stationary (unit root not rejected, KPSS r...
2	Raw hourly, seasonal difference lag 24	0.000000e+00	0.023280	0.100000	Stationary (ADF rejects unit root, KPSS does n...
3	Clear-sky index, daylight hours (concatenated)	4.586980e-23	0.830651	0.010000	Conflicting: no unit root but not level-statio...
4	Clear-sky index at 12:00 (daily)	3.983140e-08	0.600284	0.022610	Conflicting: no unit root but not level-statio...
5	Daily mean clear-sky index	3.072670e-10	0.615372	0.021239	Conflicting: no unit root but not level-statio...
6	Clear-sky index minus annual harmonic	2.090840e-25	0.026185	0.100000	Stationary (ADF rejects unit root, KPSS does n...

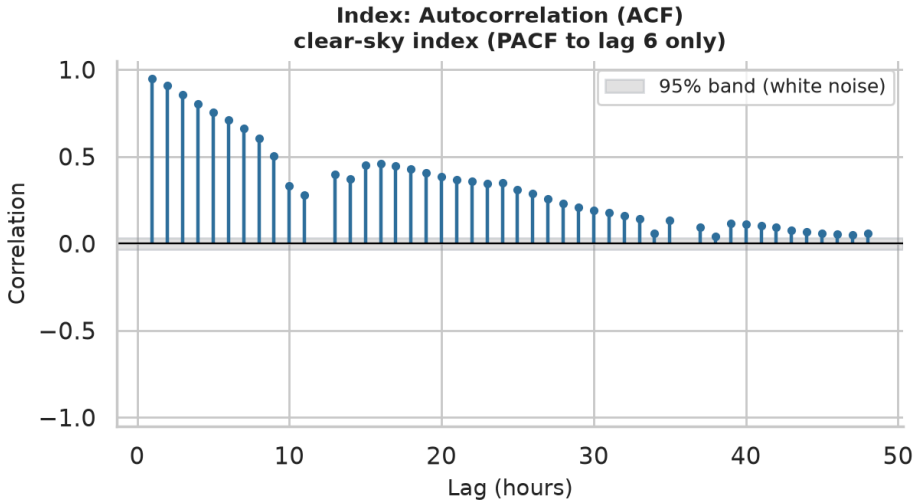
In [7]:
```
display(Image(str(config.FIG_DIR / 'fig05_rolling_statistics_and_group_effects.png')))
display(Image(str(config.FIG_DIR / 'fig06b_acf_pacf_clearsky_index.png')))
```

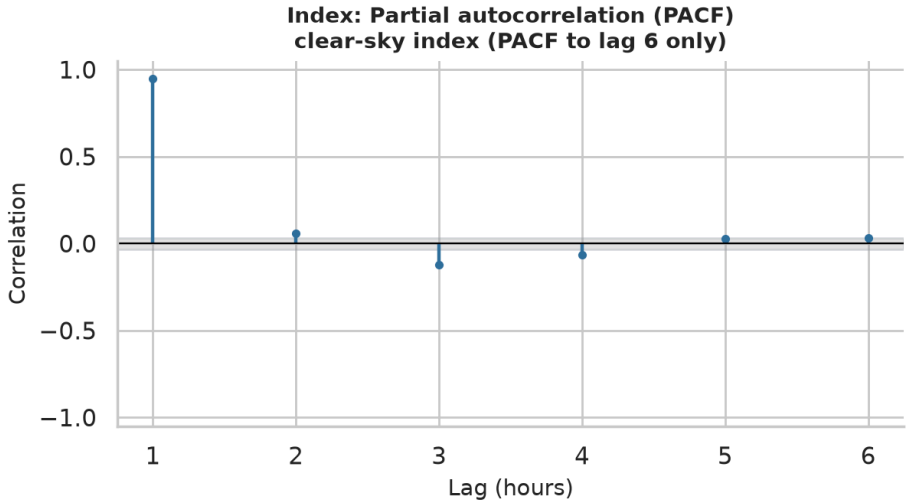

Raw output: rolling mean and standard deviation
7-day rolling mean
7-day rolling std. dev.
Power (MW)
1600
1400
1200
1000
800
600
400
2006-01
2006-03
2006-05
2006-07
2006-09
2006-11
2007-01
Date


Raw output: mean +/- std. dev. by hour of day
Power (MW)
3500
3000
2500
2000
1500
1000
500
0
0
5
10
15
20
Hour of day


Clear-sky index: daily mean and 15-day rolling mean
Index (dimensionless)
1.0
0.8
0.6
0.4
0.2
Daily mean index
15-day rolling mean
2006-01
2006-03
2006-05
2006-07
2006-09
2006-11
2007-01
Date


Clear-sky index by month (mean and spread both vary)
Index (dimensionless)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1
2
3
4
5
6
7
8
9
10
11
12
Month





4. Maximum likelihood estimation

A minimal demonstration with the Weibull model: maximise the log-likelihood numerically and compare with SciPy.

In [8]:
```
x = frame.loc[frame['in_sample'], 'clearsky_index'].to_numpy()
weibull = mm.Weibull()
fit = weibull.fit(x)
print(fit.params, fit.loglik, fit.converged)
print(mm.validate_against_scipy(x, weibull, fit))
```

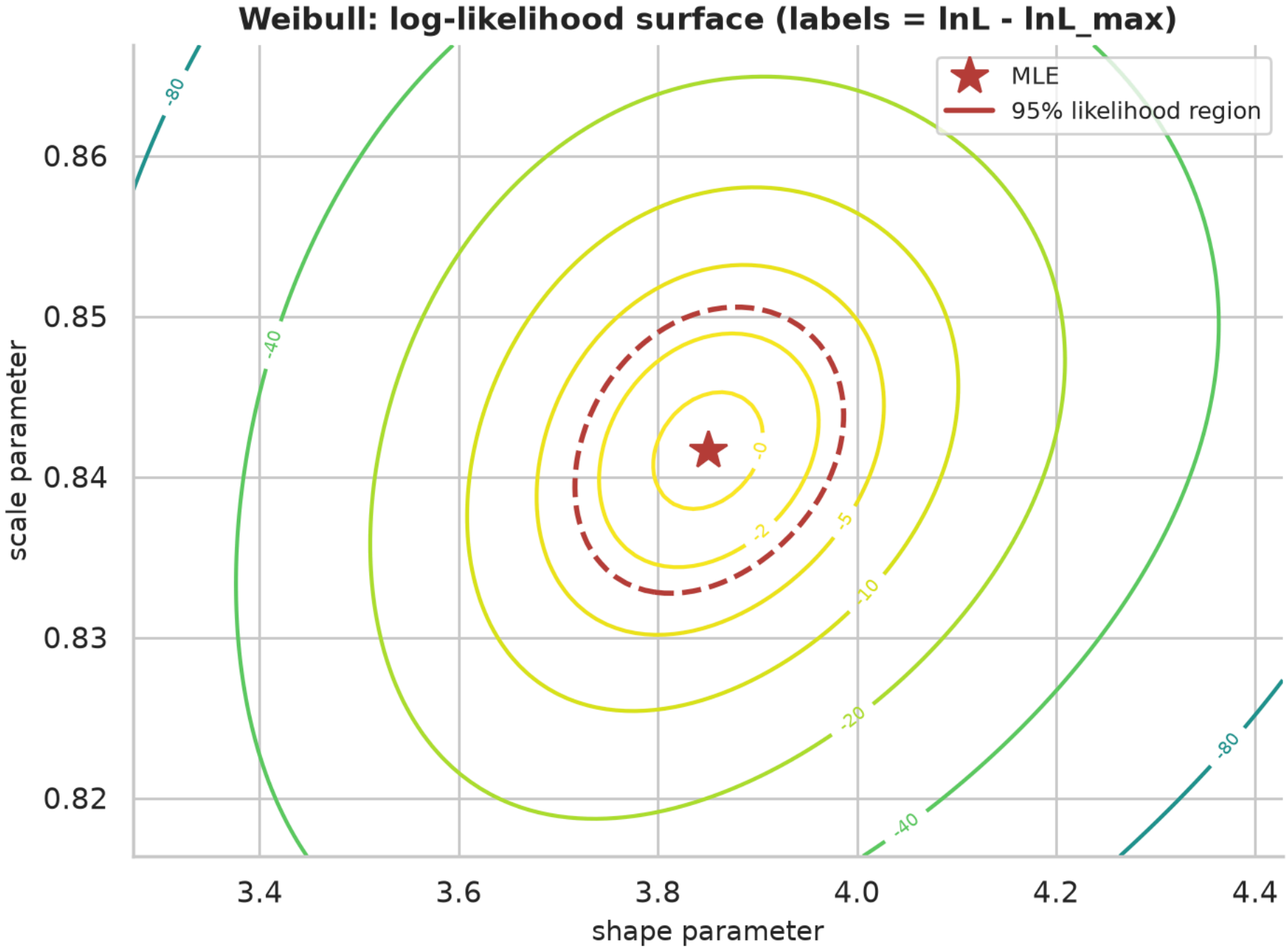
```
{'shape': 3.8502063276330274, 'scale': 0.8416354298249801} -22.06940509829809 True
{'model': 'Weibull', 'our_loglik': -22.06940509829809, 'scipy_loglik': -22.069405507743397, 'loglik_difference_ours_minus_scipy': 4.0944530610431684e-07, 'max_abs_param_difference': 4.528117677882193e-05, 'max_rel_param_difference': 1.1760576029684276e-05, 'validated': True}
```

In [9]:
```
pd.read_csv(config.RESULTS_DIR / 'mle_estimates.csv')
```

Out[9]:

	model	parameter	estimate	std_error	ci95_low	ci95_high	loglik	n
0	Normal	mu	0.763780	0.003893	0.756150	0.771410	8.79304	3846
1	Normal	sigma	0.241418	0.002753	0.236023	0.246813	8.79304	3846
2	Student-t	mu	0.812815	0.005818	0.801412	0.824218	45.12700	3846
3	Student-t	sigma	0.188409	0.005830	0.176982	0.199835	45.12700	3846
4	Student-t	nu	4.393030	0.526213	3.361650	5.424410	45.12700	3846
5	Log-normal	mu_log	-0.357052	0.008087	-0.372903	-0.341202	-1429.94000	3846
6	Log-normal	sigma_log	0.501533	0.005718	0.490324	0.512741	-1429.94000	3846
7	Gamma	shape	5.870920	0.130253	5.615630	6.126220	-789.25500	3846
8	Gamma	scale	0.130095	0.003013	0.124189	0.136001	-789.25500	3846
9	Weibull	shape	3.850210	0.055174	3.742070	3.958350	-22.06940	3846
10	Weibull	scale	0.841635	0.003639	0.834502	0.848768	-22.06940	3846
11	Beta (scaled)	alpha	3.559710	0.079257	3.404360	3.715050	91.95940	3846
12	Beta (scaled)	beta	2.718430	0.059240	2.602320	2.834540	91.95940	3846
13	2-component normal mixture	weight1	0.483891	0.014597	0.455282	0.512500	1001.05000	3846
14	2-component normal mixture	mu1	0.586236	0.008713	0.569160	0.603313	1001.05000	3846
15	2-component normal mixture	sigma1	0.234444	0.004282	0.226051	0.242837	1001.05000	3846
16	2-component normal mixture	mu2	0.930240	0.002534	0.925273	0.935207	1001.05000	3846
17	2-component normal mixture	sigma2	0.064276	0.002126	0.060110	0.068443	1001.05000	3846

In [10]:
```
display(Image(str(config.FIG_DIR / 'fig07_likelihood_surface_weibull.png')))
```


Weibull: log-likelihood surface (labels = lnL - lnL_max)
MLE
95% likelihood region
scale parameter
shape parameter
0.86
0.85
0.84
0.83
0.82
3.4
3.6
3.8
4.0
4.2
4.4
-80
-40
-0
-2
-5
-10
-20
-40
-80

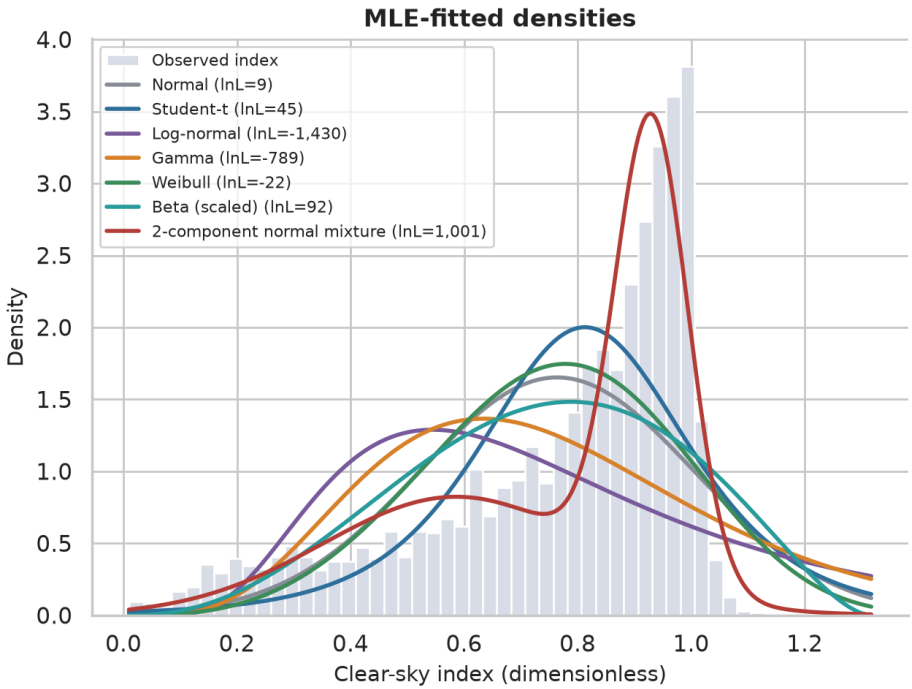
5. Model selection (AIC / BIC / goodness of fit)

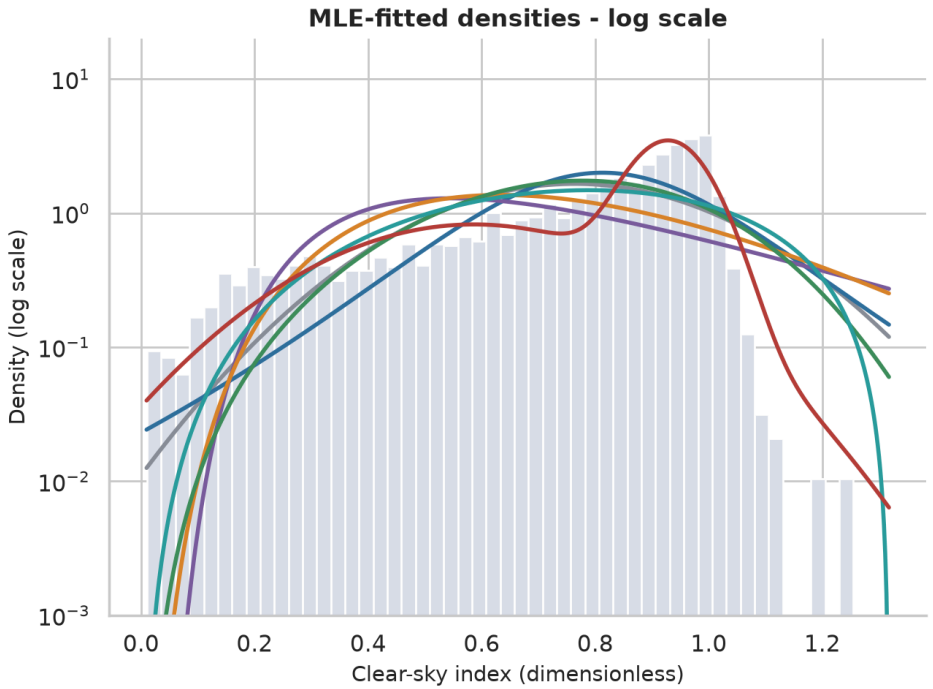
In [11]:
```
pd.read_csv(config.RESULTS_DIR / 'model_comparison.csv')[['model','k','loglik','AIC','BIC','delta_AIC','delta_BIC','we
```

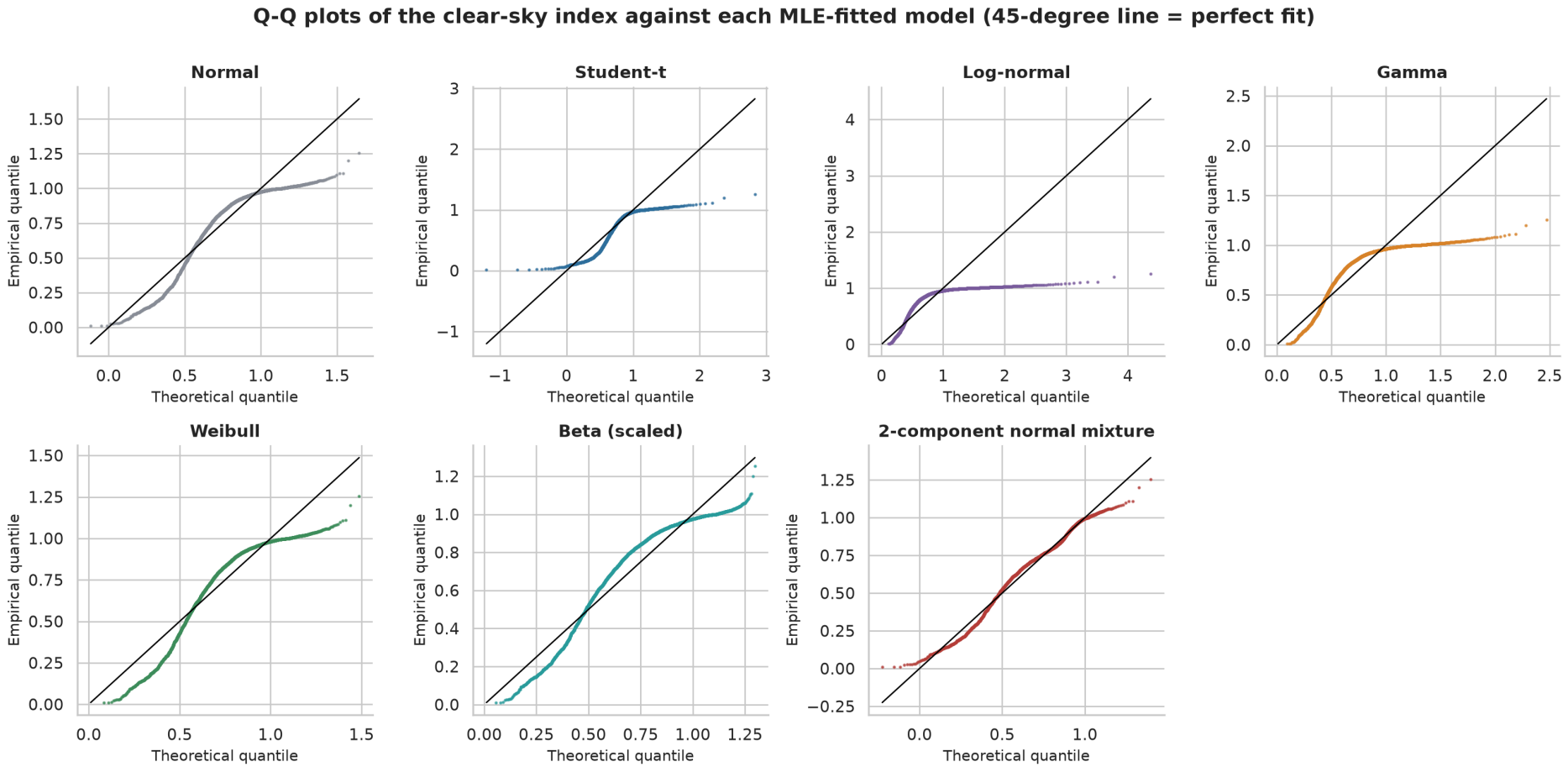
Out[11]:

	model	k	loglik	AIC	BIC	delta_AIC	delta_BIC	weight_BIC	KS_statistic	KS_p_bootstrap	AD_p_bootstrap
0	2-component normal mixture	5	1001.05000	-1992.1000	-1960.83000	0.00	0.00	1	0.036444	0.004975	0.004975
1	Beta (scaled)	2	91.95940	-179.9190	-167.40900	1812.18	1793.42	0	0.146453	0.004975	0.004975
2	Student-t	3	45.12700	-84.2541	-65.48970	1907.85	1895.34	0	0.142487	0.004975	0.004975
3	Normal	2	8.79304	-13.5861	-1.07651	1978.52	1959.75	0	0.143025	0.004975	0.004975
4	Weibull	2	-22.06940	48.1388	60.64840	2040.24	2021.48	0	0.146126	0.004975	0.004975
5	Gamma	2	-789.25500	1582.5100	1595.02000	3574.61	3555.85	0	0.184510	0.004975	0.004975
6	Log-normal	2	-1429.94000	2863.8800	2876.39000	4855.98	4837.22	0	0.206796	0.004975	0.004975

In [12]:
```
display(Image(str(config.FIG_DIR / 'fig08_fitted_densities.png')))
display(Image(str(config.FIG_DIR / 'fig09_qq_plots.png')))
```







6. Time-series models and diagnostics

In [13]:
```
pd.read_csv(config.RESULTS_DIR / 'timeseries_model_comparison.csv')[['model','k','loglik','AIC','BIC','delta_BIC','cor
```

Out[13]:

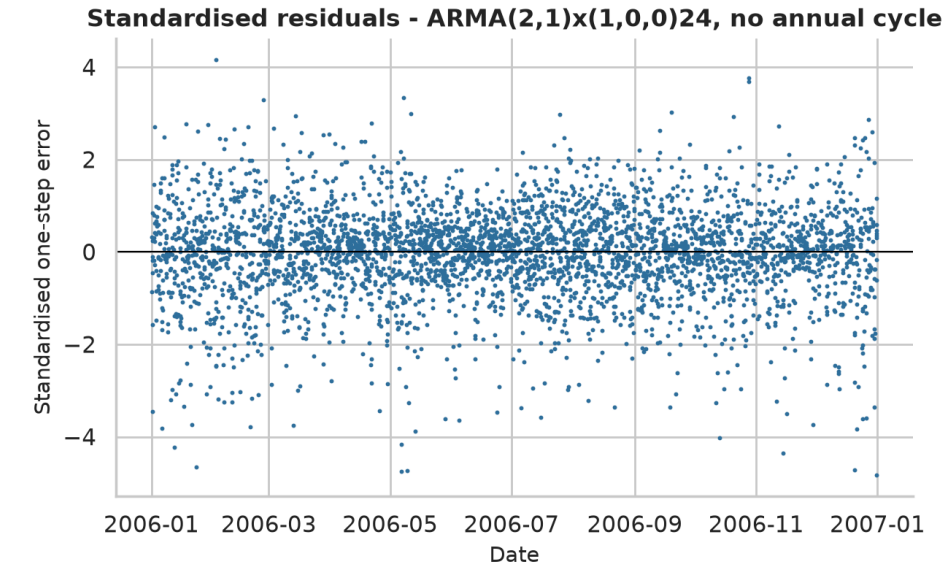
	model	k	loglik	AIC	BIC	delta_BIC	converged
0	ARMA(2,1)x(1,0,0)24, no annual cycle	6	6216.430	-12420.900	-12383.300	0.00000	True
1	ARMA(2,1)x(1,0,0)24	8	6221.680	-12427.400	-12377.300	6.01124	True
2	ARMA(1,1)x(1,0,0)24	7	6118.980	-12224.000	-12180.200	203.16400	True
3	ARMA(2,1)	7	5878.030	-11742.100	-11698.300	685.05000	True
4	ARMA(2,2)	8	5881.110	-11746.200	-11696.200	687.16200	True
5	AR(3)	7	5869.520	-11725.000	-11681.300	702.08500	True
6	AR(2)	6	5849.850	-11687.700	-11650.200	733.16200	True
7	ARMA(1,1)	6	5821.800	-11631.600	-11594.100	789.26100	True
8	AR(1)	5	5346.550	-10683.100	-10651.800	1731.51000	True
9	iid Gaussian + annual cycle	4	210.638	-413.276	-388.256	11995.10000	True

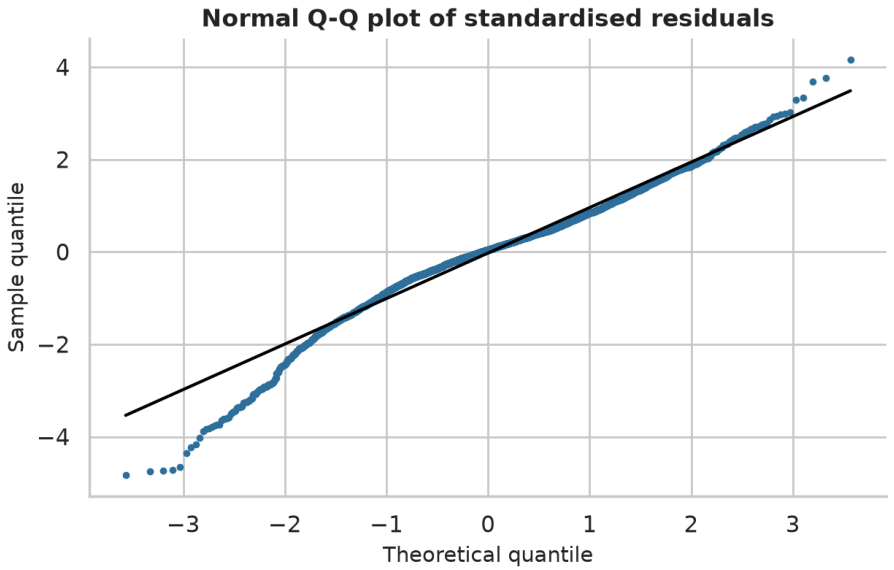
In [14]:
```
pd.read_csv(config.RESULTS_DIR / 'model_comparison_all_families.csv')
```

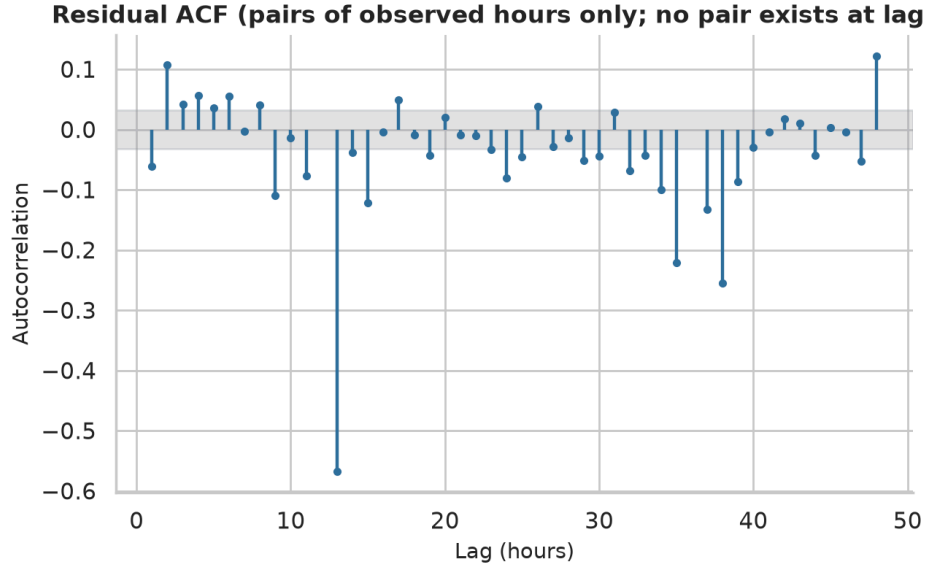
Out[14]:

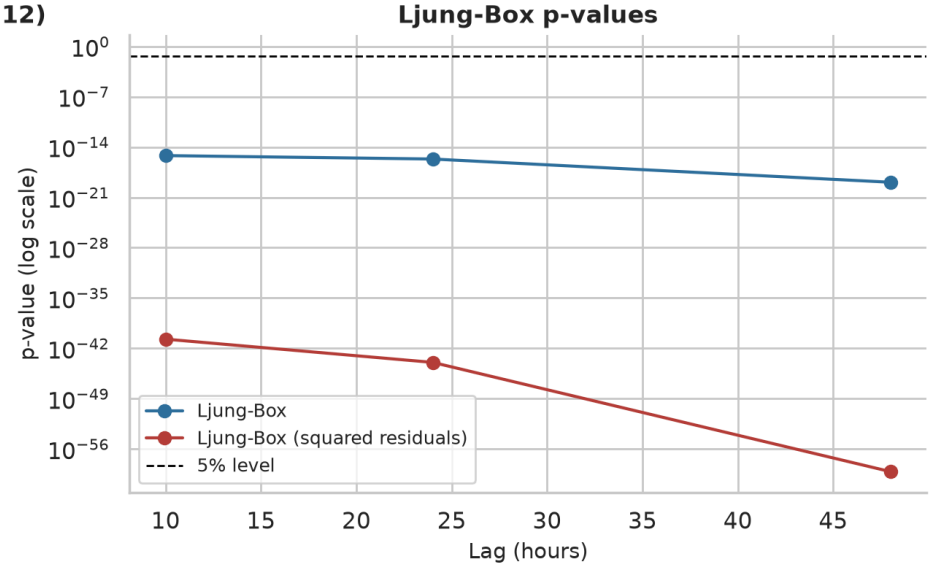
	model	k	n	loglik	AIC	BIC	family	delta_BIC
0	ARMA(2,1)x(1,0,0)24, no annual cycle	6	3846	6216.43000	-12420.9000	-12383.30000	ARMA-type (time order used)	0.00000
1	ARMA(2,1)x(1,0,0)24	8	3846	6221.68000	-12427.4000	-12377.30000	ARMA-type (time order used)	6.01124
2	ARMA(1,1)x(1,0,0)24	7	3846	6118.98000	-12224.0000	-12180.20000	ARMA-type (time order used)	203.16400
3	ARMA(2,1)	7	3846	5878.03000	-11742.1000	-11698.30000	ARMA-type (time order used)	685.05000
4	ARMA(2,2)	8	3846	5881.11000	-11746.2000	-11696.20000	ARMA-type (time order used)	687.16200
5	AR(3)	7	3846	5869.52000	-11725.0000	-11681.30000	ARMA-type (time order used)	702.08500
6	AR(2)	6	3846	5849.85000	-11687.7000	-11650.20000	ARMA-type (time order used)	733.16200
7	ARMA(1,1)	6	3846	5821.80000	-11631.6000	-11594.10000	ARMA-type (time order used)	789.26100
8	AR(1)	5	3846	5346.55000	-10683.1000	-10651.80000	ARMA-type (time order used)	1731.51000
9	2-component normal mixture	5	3846	1001.05000	-1992.1000	-1960.83000	i.i.d. distribution	10422.50000
10	iid Gaussian + annual cycle	4	3846	210.63800	-413.2760	-388.25600	ARMA-type (time order used)	11995.10000
11	Beta (scaled)	2	3846	91.95940	-179.9190	-167.40900	i.i.d. distribution	12215.90000
12	Student-t	3	3846	45.12700	-84.2541	-65.48970	i.i.d. distribution	12317.80000
13	Normal	2	3846	8.79304	-13.5861	-1.07651	i.i.d. distribution	12382.30000
14	Weibull	2	3846	-22.06940	48.1388	60.64840	i.i.d. distribution	12444.00000
15	Gamma	2	3846	-789.25500	1582.5100	1595.02000	i.i.d. distribution	13978.40000
16	Log-normal	2	3846	-1429.94000	2863.8800	2876.39000	i.i.d. distribution	15259.70000

In [15]:
```
display(Image(str(config.FIG_DIR / 'fig12_arma_residual_diagnostics.png')))
```









In [16]:
```
d = pd.read_csv(config.RESULTS_DIR / 'diagnostic_results.csv')
d[d['family'] != 'i.i.d. distribution'].query("test in ['Ljung-Box','ARCH-LM']").head(30)
```

Out[16]:

	model	test	lag	H0	statistic	p_value	reject_H0	n	family
3	iid Gaussian + annual cycle	Ljung-Box	10.0	no autocorrelation up to this lag	13546.2000	0.000000e+00	True	3846	ARMA
4	iid Gaussian + annual cycle	Ljung-Box	24.0	no autocorrelation up to this lag	14064.7000	0.000000e+00	True	3846	ARMA
5	iid Gaussian + annual cycle	Ljung-Box	48.0	no autocorrelation up to this lag	14297.4000	0.000000e+00	True	3846	ARMA
9	iid Gaussian + annual cycle	ARCH-LM	5.0	no conditional heteroskedasticity	2715.6200	0.000000e+00	True	3846	ARMA
10	iid Gaussian + annual cycle	ARCH-LM	12.0	no conditional heteroskedasticity	2718.9300	0.000000e+00	True	3846	ARMA
11	iid Gaussian + annual cycle	ARCH-LM	24.0	no conditional heteroskedasticity	2716.6700	0.000000e+00	True	3846	ARMA
15	AR(1)	Ljung-Box	10.0	no autocorrelation up to this lag	389.8300	2.023940e-78	True	3846	ARMA
16	AR(1)	Ljung-Box	24.0	no autocorrelation up to this lag	513.8290	4.663990e-94	True	3846	ARMA
17	AR(1)	Ljung-Box	48.0	no autocorrelation up to this lag	645.5340	3.767480e-106	True	3846	ARMA
21	AR(1)	ARCH-LM	5.0	no conditional heteroskedasticity	230.4910	8.395460e-48	True	3846	ARMA
22	AR(1)	ARCH-LM	12.0	no conditional heteroskedasticity	288.5190	1.204040e-54	True	3846	ARMA
23	AR(1)	ARCH-LM	24.0	no conditional heteroskedasticity	310.8340	1.097010e-51	True	3846	ARMA
27	AR(2)	Ljung-Box	10.0	no autocorrelation up to this lag	50.7896	2.880720e-08	True	3846	ARMA
28	AR(2)	Ljung-Box	24.0	no autocorrelation up to this lag	168.7400	1.302090e-24	True	3846	ARMA
29	AR(2)	Ljung-Box	48.0	no autocorrelation up to this lag	469.8190	1.353950e-71	True	3846	ARMA
33	AR(2)	ARCH-LM	5.0	no conditional heteroskedasticity	67.3284	3.682480e-13	True	3846	ARMA
34	AR(2)	ARCH-LM	12.0	no conditional heteroskedasticity	187.9220	1.005200e-33	True	3846	ARMA
35	AR(2)	ARCH-LM	24.0	no conditional heteroskedasticity	230.6160	1.108670e-35	True	3846	ARMA
39	AR(3)	Ljung-Box	10.0	no autocorrelation up to this lag	72.7765	4.051640e-13	True	3846	ARMA
40	AR(3)	Ljung-Box	24.0	no autocorrelation up to this lag	204.9890	3.770070e-32	True	3846	ARMA
41	AR(3)	Ljung-Box	48.0	no autocorrelation up to this lag	517.2590	1.627700e-81	True	3846	ARMA
45	AR(3)	ARCH-LM	5.0	no conditional heteroskedasticity	80.6010	6.282240e-16	True	3846	ARMA
46	AR(3)	ARCH-LM	12.0	no conditional heteroskedasticity	197.5510	1.043850e-35	True	3846	ARMA
47	AR(3)	ARCH-LM	24.0	no conditional heteroskedasticity	243.4380	3.286350e-38	True	3846	ARMA
51	ARMA(1,1)	Ljung-Box	10.0	no autocorrelation up to this lag	108.8360	6.605130e-20	True	3846	ARMA
52	ARMA(1,1)	Ljung-Box	24.0	no autocorrelation up to this lag	278.8160	2.350490e-46	True	3846	ARMA
53	ARMA(1,1)	Ljung-Box	48.0	no autocorrelation up to this lag	491.2020	8.159760e-76	True	3846	ARMA
57	ARMA(1,1)	ARCH-LM	5.0	no conditional heteroskedasticity	107.8940	1.141310e-21	True	3846	ARMA
58	ARMA(1,1)	ARCH-LM	12.0	no conditional heteroskedasticity	192.8330	9.802520e-35	True	3846	ARMA
59	ARMA(1,1)	ARCH-LM	24.0	no conditional heteroskedasticity	237.5180	4.847740e-37	True	3846	ARMA

7. Train/test split and out-of-sample forecasting

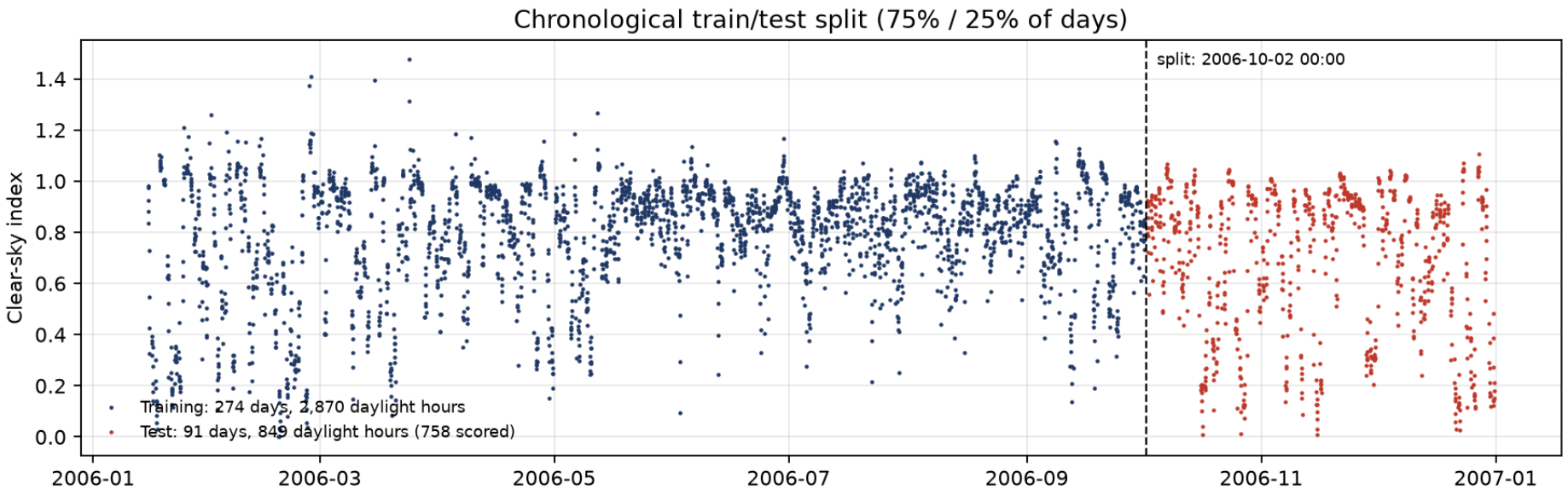
The chronological 75/25 split (first 274 days train, last 91 days test) is used only for the forecasting experiment; the distribution and ARMA/GARCH comparisons above use the whole year.

In [17]:
```
pd.read_csv(config.RESULTS_DIR / 'train_test_split.csv')
```

Out[17]:

	set	start	end	days	daylight_observations	used_for
0	Training	2006-01-01	2006-10-01	274	2870	order selection (BIC) and parameter estimation
1	Test (hold-out)	2006-10-02	2006-12-31	91	849	one-hour-ahead scoring (758 hours with an obse...

In [18]:
```
display(Image(str(config.FIG_DIR / 'fig15_train_test_split.png')))
```

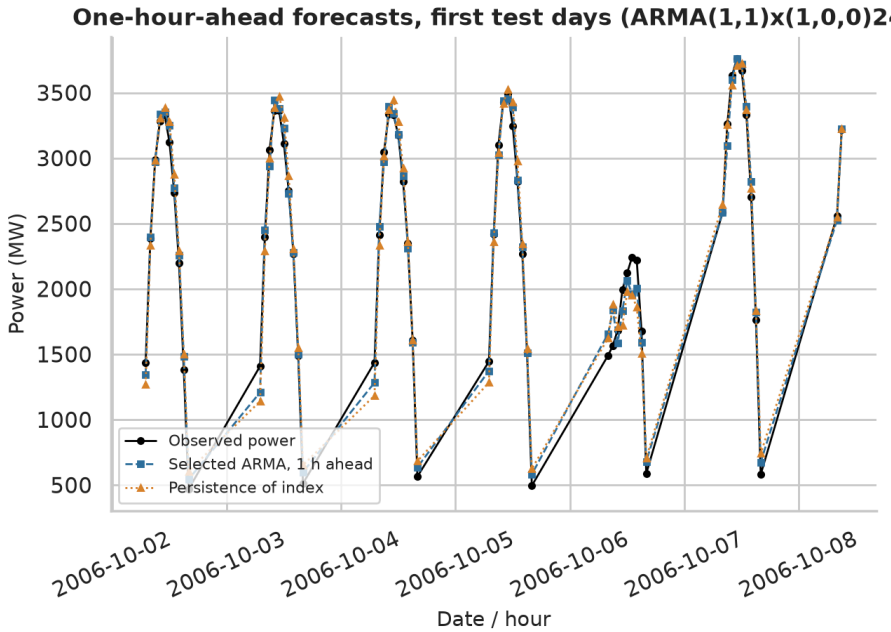


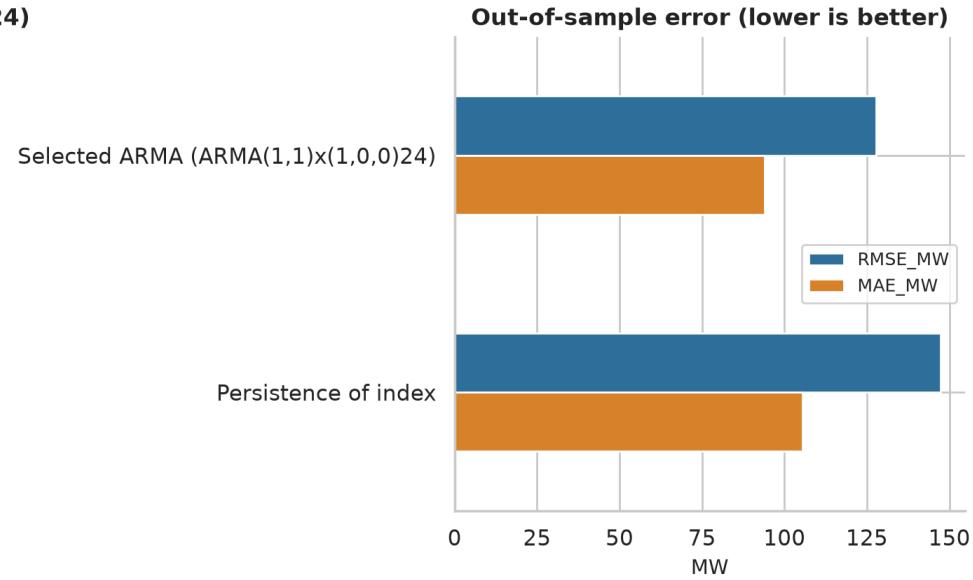
In [19]:
```
pd.read_csv(config.RESULTS_DIR / 'forecast_scores.csv')
```

Out[19]:

	method	n_test	MAE_index	RMSE_index	MAE_MW	RMSE_MW	RMSE_skill_vs_index_persistence
0	Selected ARMA (ARMA(1,1)x(1,0,0)24)	758	0.038079	0.053244	93.985	127.711	0.133524
1	Persistence of index	758	0.045597	0.068425	105.508	147.392	0.000000
2	Training-mean index	758	0.235286	0.303500	631.540	842.490	-4.715980
3	Naive power persistence (P_{t-1})	758	NaN	NaN	452.593	569.864	-2.866320

In [20]:
```
display(Image(str(config.FIG_DIR / 'fig14_forecast_evaluation.png')))
```


One-hour-ahead forecasts, first test days (ARMA(1,1)x(1,0,0)24)
Power (MW)
3500
3000
2500
2000
1500
1000
500
Observed power
Selected ARMA, 1 h ahead
Persistence of index
2006-10-02
2006-10-03
2006-10-04
2006-10-05
2006-10-06
2006-10-07
2006-10-08
Date / hour


Out-of-sample error (lower is better)
Selected ARMA (ARMA(1,1)x(1,0,0)24)
Persistence of index
RMSE_MW
MAE_MW
0
25
50
75
100
125
150
MW